Literature-implied status: approximating zone diagrams

arXiv:1002.3583 and arXiv:1208.3124

Status: literature_implied_answer (partial subcase), not an original run proof

Source questions

The source paper is Eva Kopecka, Daniel Reem, and Simeon Reich, *Zone diagrams in compact subsets of uniformly convex normed spaces*, arXiv:1002.3583, published in *Israel Journal of Mathematics* 188 (2012), 1–23 [1]. In its concluding section the paper lists several open directions: local connectedness of dominance regions, extending existence beyond the compact/uniformly convex framework, uniqueness under additional smoothness, removing the emanation-property hypothesis, and approximation of zone diagrams. The branch recorded here is the approximation branch:

> "Another interesting and natural problem is how to approximate a zone diagram in the setting described in this paper." [1, Sec. 6]

The same paragraph notes that the proof can approximate compact sites by finite subsets but gives no error estimates.

Supporting literature

The supporting paper is Daniel Reem, *On the Computation of Zone and Double Zone Diagrams*, arXiv:1208.3124, published in *Discrete & Computational Geometry* 59 (2018), 253–292 [2]. It cites the source paper as the Israel Journal of Mathematics article and treats the problem of computing zone diagrams and double zone diagrams.

The later paper defines the inner and outer approximation sequences $(I^{(n)})$ and $(O^{(n)})$ by iterating the Dom mapping [2, Algorithm 3.2]. Its main convergence theorem proves that, in a proper geodesic metric space with the geodesic inclusion property and positively separated closed sites, the limits m and M are the least and greatest double zone diagrams; in the two-site case, mixed pairs give zone diagrams [2, Theorem 4.3]. In compact worlds, the inner and outer approximations converge to these limits in Hausdorff distance [2, Corollary 4.4]. The paper also explains, in the normed-space case, how to approximate each iteration using a ray-shooting algorithm for Voronoi/dominance regions, gives practical implementation details, and records a time-complexity estimate [2, Sec. 6].

This answers a concrete subcase of the 2010 approximation question: for compact worlds inside finite-dimensional strictly convex normed spaces, and more generally compact proper geodesic spaces with the geodesic inclusion property, the inner/outer iterative procedure has a proved Hausdorff convergence target, and the normed-space iterations have an explicit approximate-computation model. When the least and greatest double zone diagrams coincide, the limit is the unique zone diagram; the later paper notes this in particular for finite dimensional Euclidean spaces [2, Corollary 4.5].

Scope limitations

This is not a full answer to every open direction in the source paper. It is a partial literature-implied answer to the approximation branch. The later paper does not give convergence-rate or iteration-count estimates: it explicitly says that estimating $n_0(\epsilon)$ remains a major open problem, and its conclusion again lists convergence-rate error estimates as open. It also does not settle the source paper's local-connectedness, all-normed-spaces/no-compactness, infinite-dimensional uniqueness, or no-emanation-property questions.

Search and evidence

The cheap run indexes had no prior row for arXiv:1002.3583, zone diagrams, or arXiv:1208.3124. The target source lines inspected were source.tex lines 765–809 of arXiv:1002.3583. The supporting source lines inspected were arXiv:1208.3124 source lines 220–224, 362–368, 437–463, 563–665, and 1234–1247. A bounded web check confirmed the publication metadata and DOI records for the source article and the supporting computation article.

References

[1] E. Kopecka, D. Reem, and S. Reich, *Zone diagrams in compact subsets of uniformly convex normed spaces*, arXiv:1002.3583; Israel J. Math. 188 (2012), 1–23. doi:10.1007/s11856-011-0094-5. `https://arxiv.org/abs/1002.3583`

[2] D. Reem, *On the Computation of Zone and Double Zone Diagrams*, arXiv:1208.3124; Discrete Comput. Geom. 59 (2018), 253–292. doi:10.1007/s00454-017-9958-8. `https://arxiv.org/abs/1208.3124`